COMSATS University Islamabad

Vehari Campus

Report

Typing Challenge Game

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Github Link: Click me For see on Github

This is my assignment project. I made a typing game with HTML, CSS, Bootstrap and JavaScript. It has two game modes. The game works on computer and mobile both.

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Registration No.	FA24-BSE-087
Program	BS Software Engineering
CLO	CLO4 — Apply the concepts of markup and scripting languages and client-side technologies
Level	Applying
My File	index.html
Demo Video	demo-video-english.mp4

1. Introduction

I made this game for my assignment. Typing is a basic skill for a software student. So I made a game that helps in typing practice and also checks speed and accuracy.

First the home screen opens. From there the player can select difficulty and game mode. Then the game starts. Both modes have their own rules and scoring. The game can be restarted and reset without reloading the page.

2. My Objectives

- To use HTML for the structure of the page.
- To use CSS and Bootstrap for design and mobile screen support.
- To use JavaScript for the game logic.
- To make two typing modes with clear rules.
- To add Easy, Medium and Hard levels that really change the game.
- To show result at the end of every game.

3. Tools I Used

Tool	What I used it for
HTML5	I made the home screen, game screen, bubbles area, keyboard and result screen with HTML.
CSS3	I made the colours, bubbles, buttons and shadows with CSS.
Bootstrap 5	I used Bootstrap for layout, cards, buttons and mobile design.
JavaScript	I wrote the game logic in simple JavaScript. I used var, simple for loops and if-else only. I did not use hard code, so I can explain it in viva.

4. Home Screen

On the home screen, first we select difficulty. For KeyMagnet, we also select the key set. Then we select a mode:

- Mode A: Bubbles — Letter Recognition. Letters come up in bubbles.
- Mode B: KeyMagnet — Keyboard Precision. A target key is shown and we have to press it.

Both cards show the name, a short description, rules and a Start button. The score rule is also shown: correct key gives +10 points and wrong key gives -5 points. Score never goes below zero.

I used one colour set for the whole website. The colours are navy, indigo, violet, pink, teal, amber and red. All screens, cards, buttons and result box use these same colours, so the website looks in one sequence.

5. Mode A: Bubbles

Rules of this mode are:

- At least 5 bubbles are shown at one time. Each bubble has one letter and it moves up.
- If we press the same letter key, the bubble is removed and we get +10 points.
- For a correct key, a green message is shown. For a wrong key, a red message is shown and 5 points are cut.
- If a bubble reaches the red Miss Line at the top, it counts as 1 miss. After 6 misses, the game is over.
- To win, we have to pop 20 bubbles on Easy, 30 on Medium and 40 on Hard.
- After every 10 pops, the game becomes faster and one more bubble is added, up to 9 bubbles.
- On mobile, we can also tap a bubble to pop it.

6. Mode B: KeyMagnet

Rules of this mode are:

- A big target key is shown on the screen with a keyboard.
- Target keys come from a JavaScript array. Key sets are: Home Row (A, S, D, F, J, K, L), Top Row (Q, W, E, R, T, Y, U, I, O, P) and All Letters (A to Z).
- Correct key gives +10 points. Wrong key gives -5 points and shows the correct key in the message.
- The game counts correct answers, wrong answers and response time. At the end it shows average time and fastest time.
- Key set and difficulty really change the game. Key set changes the keys, and difficulty changes targets and time.
- If we finish all targets before time ends, we win. If time ends, the game shows Time Up.

7. Difficulty Levels

I saved all difficulty values in one DIFF object in JavaScript. So the same difficulty is used in both modes.

Level	Bubbles	KeyMagnet
Easy	Speed 1, 5 bubbles, win at 20 pops	15 targets in 60 seconds
Medium	Speed 1.7, 6 bubbles, win at 30 pops	20 targets in 45 seconds
Hard	Speed 2.5, 7 bubbles, win at 40 pops	25 targets in 30 seconds

8. Requirements of Assignment

#	Requirement	How I did it
1	Game selection	Home screen has both modes with name, description and Start button.
2	Difficulty levels	Easy, Medium and Hard change speed, bubbles, targets and time.
3	Gameplay controls	Start, Restart and Reset buttons work without reloading the page.
4	Score and status	Score, Correct, Wrong, Misses Left / Time Left, difficulty and progress bar are always shown.
5	Immediate feedback	Green message for correct and red message for wrong. In KeyMagnet, the pressed key also flashes green or red.
6	Game completion	Bubbles: Win or Game Over after 6 misses. KeyMagnet:

		Win or Time Up.
7	Results screen	At the end, score, correct, wrong, accuracy and mode stats are shown.
8	State reset	Restart resets score, timer, misses, targets and bubbles.
9	Timer management	stopTimers() stops all old timers before a new game and at the end, so two timers never run together.
10	Responsive design	Bootstrap grid and keyboard work on desktop and mobile. I also added screen sizes 768px and 480px for small screens.

9. Scoring

Action	Points / Result
Correct key (both modes)	+10 points
Wrong key (both modes)	-5 points, score never below 0
Bubble reaches miss line	1 miss
6 misses in Bubbles	Game Over
Timer ends in KeyMagnet	Time Up

10. How to Run My Game

- Open index-simple.html in a browser like Chrome. No server or setup is needed.
- Select Difficulty on the home screen. For KeyMagnet, also select Key Set.
- Press Start for Bubbles or KeyMagnet and play with the keyboard.
- Use Restart to play again, or Reset / Home to go back, without reloading the page.
- Internet is needed only for Bootstrap CSS. All game logic runs in the browser.

11. Testing I Did

Test	Expected result	Status
Press a matching bubble letter	Bubble removed, score +10, green message	Pass
Press a wrong letter	Score -5, red message	Pass
Bubble crosses the miss line	Miss counted, new bubble comes	Pass
6 misses in Bubbles	Game Over result screen	Pass
Pop the target number of bubbles	Win result screen	Pass
Press the correct target key in KeyMagnet	Score +10, next target shown	Pass
Timer ends in KeyMagnet	Time Up result screen	Pass
Restart during a game	Score and timer reset, old timer does not continue	Pass
Reset during a game	Home screen comes back, game stops	Pass
Play on a mobile-size screen	Layout, buttons and keyboard work well	Pass

12. Demo Video

My demo video is a screen recording of my real game. In the video, first the home screen is shown. Then a full Bubbles game is played and it ends in a Win screen with Score 200, Popped 20, Misses 0 and Accuracy 100%. Then I go back to home and play a full KeyMagnet game. It ends in a Win screen with Score 55, Correct 8, Wrong 7, Accuracy 53%, average time 765 ms and fastest time 30 ms. This video shows both modes fully, as required.

13. Conclusion

In this project I made two complete typing modes. I added difficulty levels, Start, Restart and Reset buttons, scoring, feedback, win and game over conditions, result screens and mobile design. I kept the code simple, so I can explain every part in viva: HTML structure, CSS and Bootstrap design, and JavaScript logic.

Appendix: Main JavaScript Functions

- startGame(m) — It resets everything, stops old timers, and starts the selected mode.
- stopTimers() — It stops all timers, so no old timer keeps running.
- addBubble() / moveBubbles() / popBubble(b) — These make bubbles, move them up, and remove them in Mode A.
- buildKeyboard() / nextTarget() / handleKey(key) — These make the keyboard, show the next target, and check the key in Mode B.
- averageTime() — It finds average response time with a simple for loop.
- updateStatus() — It updates score, correct, wrong, misses/time left and progress.
- restartGame() / resetGame() — Restart starts the game again, and Reset goes back to Home.
- endGame(title) — It stops the game and shows the result screen.